



NPTEL Online Certification Course
Applied Accelerated Artificial Architecture
Assignment 3 : Detailed Solution
Indian Institute of Technology Guwahati

Question 1

In Container Orchestration, which phase of the orchestration process involves continuously comparing the user's declared target configuration (desired state) against what is currently running (actual state)?

- A) You declare
- B) IT compiles
- C) It reconciles
- D) It watches

Correct Answer: D) It watches

Explanation: The "watch" phase continuously monitors the current cluster state to detect drift from the desired state. This streaming observation triggers reconciliation loops to restore target configurations automatically.

Question 2

Which internal Google system served as the original foundation for Kubernetes before it was open-sourced in 2014 and donated to the CNCF in 2015?

- A) Kuberflow
- B) Helm
- C) Borg
- D) Argo

Correct Answer: C) Borg

Explanation: Borg was Google's internal large-scale cluster management system built to manage millions of jobs. Kubernetes was designed as an open-source rewrite of Borg, drawing on years of operational lessons learned.

Question 3

In the Kubernetes control plane architecture, which component acts as "the only door in" by handling authentication, validation, and admission control?

- A) etcd
- B) API Server

- C) Scheduler
- D) Controller Manager

Correct Answer: B) API Server

Explanation: The `kube-apiserver` acts as the central hub and primary entry point for all control plane interactions. It authenticates, authorizes, and validates all incoming REST requests before storing state changes in `etcd`.

Question 4

Which component on a Kubernetes Worker Node is responsible for configuring iptables or eBPF rules to ensure network traffic for Services properly routes to live Pod IPs?

- A) kubelet
- B) kube-proxy
- C) containerd
- D) NVIDIA device plugin

Correct Answer: B) kube-proxy

Explanation: `kube-proxy` runs on every node to manage network routing rules via technologies like `iptables` or IPVS/eBPF. It proxies virtual Service IP traffic directly to the dynamically changing underlying Pod network endpoints.

Question 5

Which of the following correctly describes how storage volumes and IP addresses are assigned to containers within a Pod?

- A) Each container gets its own IP address, but volumes cannot be shared across containers.
- B) A single Pod IP is shared by all containers in the Pod, and volumes can mount into any container in the Pod.
- C) Volumes mount exclusively to the main application container and are inaccessible to sidecar containers.
- D) IP addresses are assigned at the container level, while storage volumes are assigned at the node level.

Correct Answer: B) A single Pod IP is shared by all containers in the Pod, and volumes can mount into any container in the Pod.

Explanation: Pods serve as the atomic unit of scheduling where all co-located containers share a single network namespace (and IP). Mounted storage volumes are shared across containers in the same Pod via local mount points.

Question 6

Which Kubernetes component is required to advertise a GPU as an extended resource to the scheduler?

- A) Docker Desktop Virtual Machine
- B) Nvidia Device Plug-in
- C) WSL2 Ubuntu Host
- D) Cube Control Labeling Module

Correct Answer: B) Nvidia Device Plug-in

Explanation: The NVIDIA Device Plugin is a DaemonSet that discovers host GPU hardware and registers GPU capacity with `kubelet`. This enables the Kubernetes scheduler to track and allocate GPUs as extended resources ([\[nvidia.com/gpu\]](https://nvidia.com/gpu)(<https://nvidia.com/gpu>)).

Question 7

What information is explicitly displayed inside the pod logs when running the `gpu-smoke-working.yaml` program successfully?

- A) A benchmark script testing floating-point operations
- B) An `nvidia-smi` table format output
- C) A direct memory dump of the physical GPU VRAM
- D) A continuous stream of CUDA core temperature metrics

Correct Answer: B) An `nvidia-smi` table format output

Explanation: Smoke tests for GPU access execute a lightweight container invoking `nvidia-smi` to verify hardware access. Success produces the standard tabular output listing GPU model, driver details, and CUDA runtime availability.

Question 8

Why are headless services configured specifically for stateful AI clusters instead of utilizing a default cluster IP address?

- A) They act as an encrypted cluster firewall layer.
- B) They map DNS lookups directly to individual pod IPs.
- C) They load-balance external network data globally.
- D) They alter the core container base settings dynamically.

Correct Answer: B) They map DNS lookups directly to individual pod IPs.

Explanation: Headless Services (specifying `clusterIP: None`) bypass single virtual IP load balancing. They return direct `A/AAAA` DNS records for individual Pod IPs, enabling direct point-to-point worker communication in stateful setups.

Question 9

Which operational practice must be followed when managing sensitive access tokens like Hugging Face (HF) API keys?

- A) Store the tokens directly within plain-text configuration maps.
- B) Bake authorization keys straight into the Dockerfile source layer.
- C) Isolate and hold the sensitive data using Kubernetes Secrets.
- D) Host credentials on a public cloud load-balancing endpoint.

Correct Answer: C) Isolate and hold the sensitive data using Kubernetes Secrets.

Explanation: Hardcoding sensitive tokens into images or ConfigMaps exposes them to leakages and source control risks. Kubernetes Secrets decouple sensitive credentials, allowing secure mounting at runtime without exposing raw keys.

Question 10

Which directive is used within a container pod's manifest file to import all key-value configurations directly into the container's environment?

- A) dataFrom
- B) configMapRef
- C) envFrom
- D) valueFrom

Correct Answer: C) envFrom

Explanation: `envFrom` bulk-injects all key-value pairs from a specified ConfigMap or Secret as container environment variables. This avoids explicitly declaring individual variables using `valueFrom`.

Question 11

If a pod manifest references a Secret named `ml-tokens` via `envFrom`, what occurs if the `create-secret.sh` script has not been run first?

- A) The pod skips the variables and starts anyway.
- B) The cluster dynamically generates empty values.
- C) The pod gets stuck in a creation error state.
- D) The system switches the type to a ConfigMap.

Correct Answer: C) The pod gets stuck in a creation error state.

Explanation: When a Pod references a non-existent Secret or ConfigMap during startup, `kubelet` cannot populate required environment variables. As a result, the Pod fails to start and remains stuck in `CreateContainerConfigError`.

Question 12

During the initial phase of the Canary rollout demonstration, how is the traffic split configured between the stable version and the canary version?

- A) 50% Stable / 50% Canary
- B) 75% Stable / 25% Canary
- C) 90% Stable / 10% Canary
- D) 100% Stable / 0% Canary

Correct Answer: C) 90% Stable / 10% Canary

Explanation: Canary rollouts mitigate risk by initially exposing new releases to a minimal user slice. A 90/10 split allows safely observing metrics and error logs on 10% canary traffic before committing to a full deployment.

Question 13

What happens to the cluster IP mapping when all worker pods are clubbed together into a regular Virtual IP (VIP) service?

- A) A single, unified Cluster IP is generated for the service.
- B) Each pod continues to use a completely unmapped public IP address.
- C) The control plane crashes due to an IP conflict.
- D) The service shifts exclusively to using external NodePorts.

Correct Answer: A) A single, unified Cluster IP is generated for the service.

Explanation: Standard Kubernetes Services assign an internal virtual IP (ClusterIP) that abstracts underlying Pod pods. Traffic directed to this single IP is automatically load-balanced across all matching endpoints.

Question 14

When verifying the injected configuration inside the training pod, which command structure is used by the speaker to check the active environment variables?

- A) `kubectrl env-verify pod training-pod`
- B) `kubectrl exec <pod-name> -- env`
- C) `kubectrl inspect config --pod`
- D) `kubectrl get env-form training-pod`

Correct Answer: B) `kubectrl exec <pod-name> -- env`

Explanation: Executing `kubectrl exec <pod-name> -- env` runs the Linux `env` command directly inside a running container. This dumps all active environment variables, validating that ConfigMaps or Secrets were successfully injected.

Question 15

What is generated for three stateful set workers when configuring a Headless Service?

- A) A single virtual IP address.
- B) Separate, individual IP addresses for each worker pod.
- C) A public load balancer endpoint.
- D) An egress gateway routing table.

Correct Answer: B) Separate, individual IP addresses for each worker pod.

Explanation: Headless Services forgo allocating a single shared virtual IP. Instead, DNS query responses provide direct, individual IP addresses for each matching Pod in the StatefulSet to support direct inter-node communication.